

Blender's Simulation Nodes

A workshop on creating a melting effect with Geometry Nodes in Blender

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ABSTRACT

Geometry Nodes is a procedural modeling and animation system that has been part of Blender since 2021, with Blender 2.92. It was initially focused on set dressing, procedural modeling and initial hair grooming. However in Blender 3.6 simulation nodes were finally added. The class will be a hands-on show and tell of Geometry Nodes with particular emphasis on the brand new simulation systems.

CCS CONCEPTS

• **Computing methodologies** → *Simulation types and techniques; Simulation tools; Physical simulation.*

KEYWORDS

Procedural modeling, Blender, Geometry Nodes, Simulation Nodes, simulation nodal workflow, workshop

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1 INTRODUCTION

At the end of the course artists should understand the principles behind geometry nodes and the simulation nodes, as well as have some hands-on experience on creating a simple melting effect simulation.

1.1 Requirements

This course is intended for beginner Blender artists. No previous knowledge of Geometry Nodes is required, but understanding of the principles of Blender will help.

Audience participation is encouraged, to ask questions and clarifications on the topics discussed. One of the speakers will be leading the workshop, while the other will follow attendee's progress and provide help. This requires Blender 3.6 LTS.

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Figure 1: Geometry Nodes explosions in Charge. © Blender Foundation | blender.org

1.2 Outline

- (1) Introduction to Geometry Nodes
- (2) Animation vs Simulation
- (3) Propagate Data of Simulation Steps
- (4) Mesh Processing in Geometry Nodes
- (5) Using Procedural Data in the Shader
- (6) Practical Example: Melting Effect

2 ANIMATION VS SIMULATION

The workshop will cover the difference between animation and simulation while exploring freely available example files.

Geometry Nodes has focused on animation so far. In those cases each frame can be calculated independently of any other, and no “state” carries over. Simulation on the other hand is about using the result of one frame to influence the next one. That way even a set of simple rules can lead to complex results with the passing of time. The most common type of them is physics simulation, with specific solvers for physical phenomena.

In the first section, the fundamentals of simulation with geometry nodes will be explained and illustrated with simple examples, such as the Game of Life. From there a more complex melting simulation setup will be taught and explained step by step for the audience to follow along.

3 CONCLUSION

At the end of this hands-on lab artists should have a better understanding of Geometry Nodes while dipping their toes in simulation. The topic is not exhausted here, but the foundation will help them

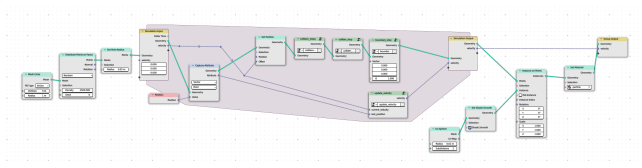


Figure 2: Node-tree example, the simulation zone is shown in light purple

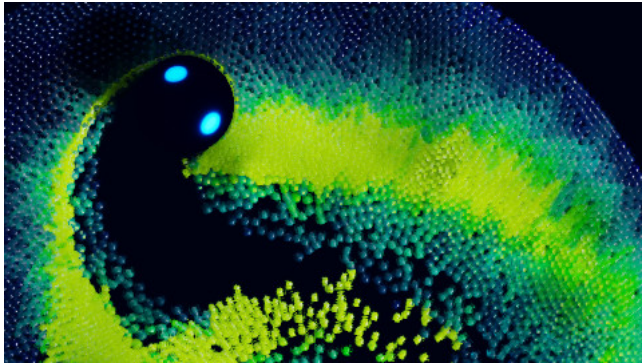


Figure 3: Index of Nearest demo CC-BY Sean Christofferson

to understand the possibilities of the system and pursue further training.

4 LINKS

Additional material can be found online:

- Simulation Nodes demo files: <https://www.blender.org/download/demo-files/#simulation>
- Simulation Nodes design, Geometry Nodes workshop 2022: <https://code.blender.org/2022/11/geometry-nodes-workshop-2022/#simulation-nodes>
- "We can do that with Geometry Nodes", live tutorial by Simon Thommes at Blender Conference 2022: <https://www.youtube.com/watch?v=1nvzwhbL-k0>
- "Geometry Nodes from Scratch", geometry nodes course by Simon Thommes at the Blender Studio platform: <https://studio.blender.org/training/geometry-nodes-from-scratch/>